

Weekly Diary – Week 11

Internship Organization: Vikaspedia

Duration: 02/04/2026 – 07/04/2026

Objective of the Week

The main objective for this week was to develop a clear understanding of browser fingerprinting and explore various techniques used to identify and track guest users across multiple sessions without requiring authentication. The focus was on studying how unique identifiers can be generated using browser-specific attributes and analyzing their applications in analytics, personalization, and security-related scenarios.

Date – 02 April

The week began with a detailed study of the project requirements related to guest user fingerprinting. The importance of identifying unauthenticated users was explored, particularly in areas such as user behavior analysis, personalized content delivery, and fraud prevention. Additionally, the fundamental working principles of browser-based tracking systems were reviewed to build a strong conceptual foundation.

Date – 03 April

On this day, in-depth research was conducted on browser fingerprinting techniques. Various browser attributes, including user agent, screen resolution, time zone, and language preferences, were examined to understand how they contribute to creating a unique identifier for each user. The process of combining these attributes to generate a consistent fingerprint was also studied.

Date – 04 April

Different fingerprinting libraries such as FingerprintJS and ClientJS were explored. A comparison was made based on their implementation approaches, ease of integration into web applications, and their effectiveness in generating stable and reliable identifiers across sessions.

Date – 06 April

Advanced fingerprinting techniques were analyzed, including canvas-based fingerprinting and WebGL-based identification methods. These approaches were studied to understand how graphical rendering differences across devices and browsers can be used to enhance the uniqueness of generated fingerprints, thereby improving tracking accuracy.

Date – 07 April

The final day focused on understanding the privacy concerns and legal considerations associated with browser fingerprinting. Regulations such as General Data Protection Regulation (GDPR) were studied to understand compliance requirements. Emphasis was placed on obtaining user consent and ensuring ethical handling and storage of tracking data to maintain transparency and trust.